Technology Stack

Date	15 March 2026
Team ID	unm293200909
Project Name	FitBuddy - AI Fitness Plan Generator using Gemini Models
Name	Kavyasri.A
Maximum Marks	2 Marks

Define Your Technology Stack:

FitBuddy is a FastAPI web application that generates personalised 7-day workout plans, nutrition or recovery tips, and feedback-based plan updates using Google Gemini models, with user data stored in SQLite.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose	Work Assigned to Kavyasri.A
1	**Frontend / Client-Side**	HTML, CSS and Jinja2 templates (index.html, result.html, all_users.html)	Server-rendered templates integrate directly with FastAPI, need no separate build step, and give a clean, simple form-based interface for users and the admin dashboard.	Design and build index.html, result.html and all_users.html with Jinja2.
2	**Backend / Server-Side**	Python with FastAPI, served by Uvicorn (ASGI)	FastAPI is fast, has simple routing and validation, and provides both REST endpoints and HTML responses. Python also works smoothly with the Gemini SDK.	Develop FastAPI app, routes.py logic and Uvicorn server setup.
3	**Database / Data Storage**	SQLite with SQLAlchemy ORM	Lightweight, file-based and zero-configuration, ideal for storing user details, generated plans and updated plans locally. SQLAlchemy keeps data access clean and easy to maintain.	Create SQLAlchemy models and store users and plans in SQLite.
4	**Cloud / Hosting / Deployment**	Local deployment using Uvicorn in a Python virtual environment (venv)	Suits the project scope: easy to run and test with uvicorn app.main:app --reload. The app can later be moved to a cloud platform such as Render or Docker if needed.	Run and test the app locally; verify all four scenarios.
5	**Version Control & CI/CD**	Git and GitHub	Tracks code changes, supports team collaboration, and keeps a backup of the project. Branching lets members work on separate milestones safely.	Create the GitHub repository and commit code milestone by milestone.

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6	**Third-Party APIs / Other Tools**	Google Gemini API: Gemini 1.5 Pro and Gemini Flash (google-generativeai)	Gemini 1.5 Pro generates and updates structured 7-day workout plans from user feedback; Gemini Flash quickly generates short nutrition and recovery tips. Both are API-based, so no GPU or model hosting is required.	Set up the Gemini API key and write prompts for plans, updates and tips.